Results

Candidate Outcome I

The generated loop selected `exp-mlp-tanh-64` (`One-hidden-layer tanh MLP`, `MLP`) after evaluating 4 candidate(s) from a proposed set of 5. The `nearest_centroid_baseline` (`nearest-centroid`) baseline reached `82.6%` `test_accuracy`, while the selected candidate reached `89.4%`, an absolute change of `6.8%`. The selected model has 50890 parameters. The transformer-candidate evaluated flag is `true`, and the candidate budget exhausted flag is `true`, which means the ledger records `1` deferred proposal(s) rather than expanding the run automatically.

Candidate Outcome II

The benchmark score is `1`. That score is not a model-quality claim by itself; it is a compact grading artifact for the methods contract: metric improvement, budget compliance, offline execution, and selected-candidate recording. Rank-stability diagnostics report that the selected candidate is top ranked in `72.5%` of deterministic bootstrap resamples, with runner-up `exp-mlp-relu-32`. The candidate score figure is registered as @fig:ml_candidate_scores, the confusion matrix as @fig:ml_confusion_matrix, the per-class diagnostic as @fig:ml_per_class_accuracy, the learning curves as @fig:ml_learning_curves, the complexity diagnostic as @fig:ml_complexity_accuracy, the selected-candidate error examples as @fig:mnist_error_examples, the candidate lifecycle diagnostic as @fig:autoresearch_candidate_lifecycle, rank stability as @fig:ml_candidate_rank_stability, the training-dynamics diagnostic as @fig:ml_training_dynamics, the final readiness matrix as @fig:autoresearch_stage_matrix, and the process closure as @fig:autoresearch_closure_flow.

Candidate Outcome III

Table 1: Candidate accuracy intervals from `output/data/ml_candidate_intervals.json`; intervals describe the fixed local test split.

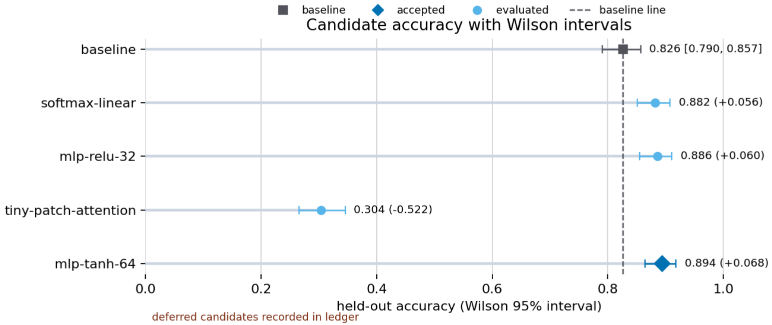
Candidate	Status	Correct/test	Accuracy	Wilson 95% interval
baseline	baseline	413/500	82.6%	79.0% to 85.7%
softmax linear	rejected	441/500	88.2%	85.1% to 90.7%
mlp relu 32	rejected	443/500	88.6%	85.5% to 91.1%
tiny patch attention	rejected	152/500	30.4%	26.5% to 34.6%
mlp tanh 64	accepted	447/500	89.4%	86.4% to 91.8%

Candidate Outcome IV

Table 2: Candidate rank-stability table from `output/data/ml_candidate_rank_stability.json`; frequencies are deterministic local bootstrap summaries.

Candidate	Observed rank	Top-rank frequency	Mean rank	Accuracy
softmax linear	3	4.2%	2.554	88.2%
mlp relu 32	2	23.3%	2.130	88.6%
tiny patch attention	4	0.0%	4.000	30.4%
mlp tanh 64	1	72.5%	1.316	89.4%

Run-Derived Figures I



Run-Derived Figures II

Figure 1: Held-out accuracy with Wilson intervals for the baseline and evaluated candidates from output/data/ml_candidate_intervals.json; accepted candidate exp-mlp-tanh-64 improves accuracy from 82.6% to 89.4% on the fixed subset, with deferred proposals kept in the candidate ledger. Generation method: Lollipop accuracy comparison with Wilson interval error bars and direct labels. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Run-Derived Figures III

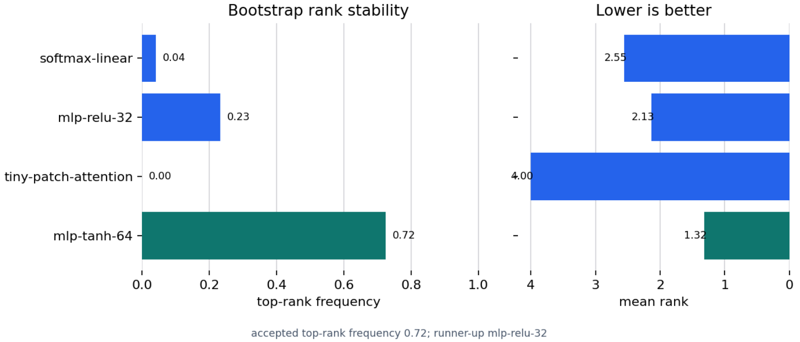
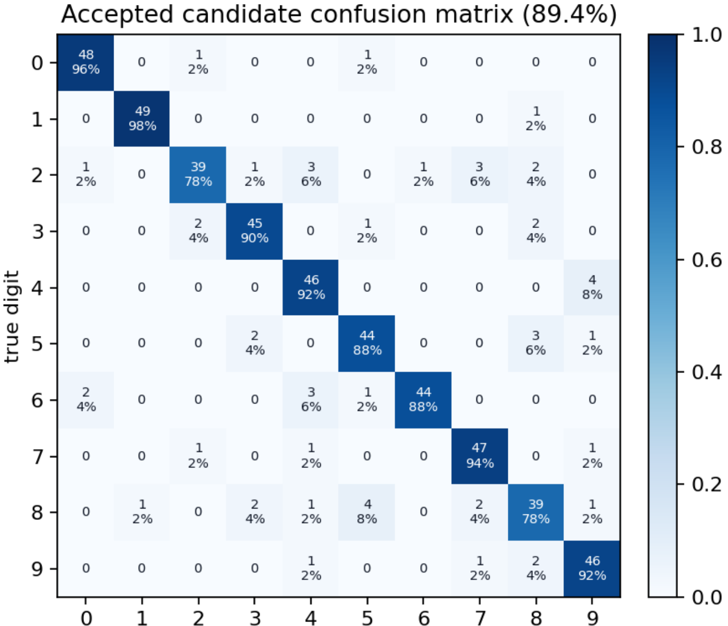
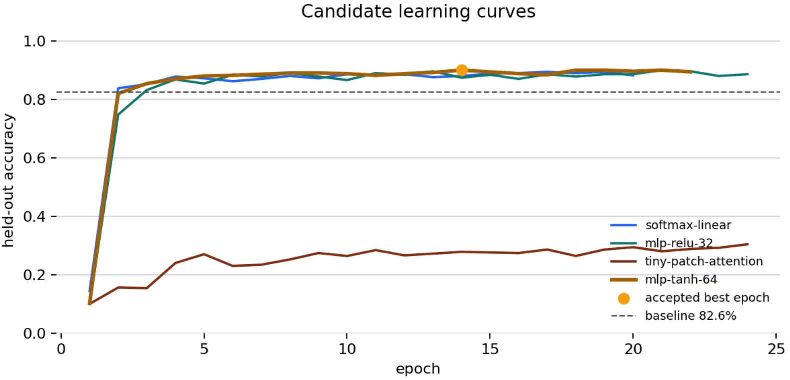


Figure 2: Rank-stability summary for exp-mlp-tanh-64 from output/data/ml_candidate_rank_stability.json; deterministic local bootstrap resampling shows how often each evaluated candidate ranks first under the fixed test split. Generation method: Bootstrap top-rank frequency and mean-rank comparison. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Run-Derived Figures IV

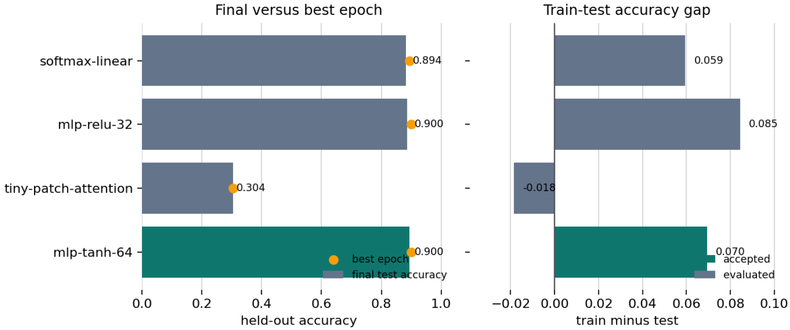


Training And Error Diagnostics I



Training And Error Diagnostics II

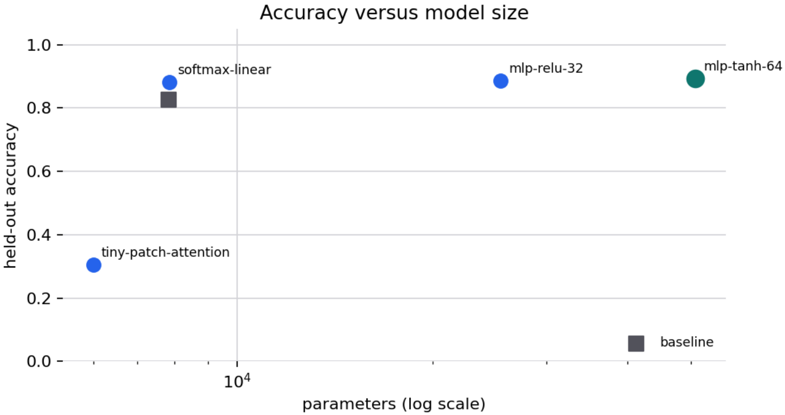
Figure 5: Epoch-level held-out accuracy curves for evaluated candidates from output/data/ml_training_history.csv; the accepted curve is visually emphasized for exp-mlp-tanh-64. Generation method: Epoch-level held-out accuracy lines with accepted best-epoch marker. Registry metadata records the generation method, source artifact, and claim boundary for validation.



Training And Error Diagnostics III

Figure 6: Configured-training dynamics for evaluated candidates from output/data/ml_training_diagnostics.json; exp-mlp-tanh-64 is highlighted while best-epoch markers and train-test gaps remain bounded to the local run. Generation method: Final and best-epoch accuracy bars plus train-test gap bars. Registry metadata records the generation method, source artifact, and claim boundary for validation.

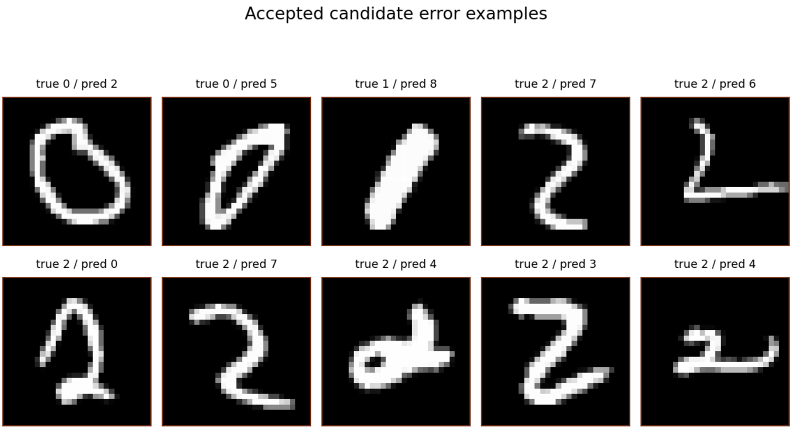
Training And Error Diagnostics IV



Training And Error Diagnostics V

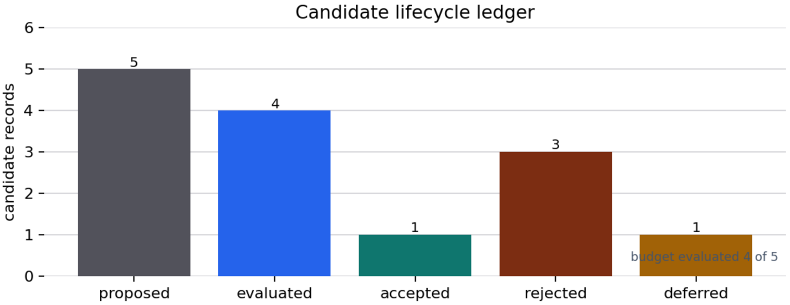
Figure 7: Parameter-count versus held-out accuracy for the baseline and evaluated candidates from output/data/ml_task_results.json; the accepted candidate is highlighted without claiming a general scaling law. Generation method: Log-parameter scatter plot against held-out accuracy. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Training And Error Diagnostics VI



Training And Error Diagnostics VII

Figure 8: First accepted-candidate error examples for exp-mlp-tanh-64, sourced from output/data/ml_error_examples.json and data/mnist_small.npz; these images support qualitative diagnosis only. Generation method: Deterministic grid of the first accepted-candidate misclassifications. Registry metadata records the generation method, source artifact, and claim boundary for validation.



Training And Error Diagnostics VIII

Figure 9: Candidate lifecycle ledger from output/data/ml_candidate_ledger.json: 4 evaluated out of 5 proposed candidates, with deferred proposals kept visible instead of executed automatically. Generation method: Candidate lifecycle status-count bar chart. Registry metadata records the generation method, source artifact, and claim boundary for validation.

The selected candidate's best held-out epoch is `14`, its final learning rate is `0.144`, its train-loss reduction is `4.161`, and its final train-test accuracy gap is `7.0%`. These values come from the configured training diagnostics rather than from a manually maintained result summary.

Table 3: Configured-training diagnostics from `output/data/ml_training_diagnostics.json`.

Candidate	Status	Best epoch	Best test accuracy	Final test accuracy	Train-test gap	Loss reduction	Final learning rate
softmax linear	rejected	17	89.4%	88.2%	5.9%	2.917	0.273

Training And Error Diagnostics IX

mlp relu 32	rejected	21	90.0%	88.6%	8.5%	4.341	0.178
tiny patch attention	rejected	24	30.4%	30.4%	-1.8%	0.165	0.214
mlp tanh 64	accepted	14	90.0%	89.4%	7.0%	4.161	0.144

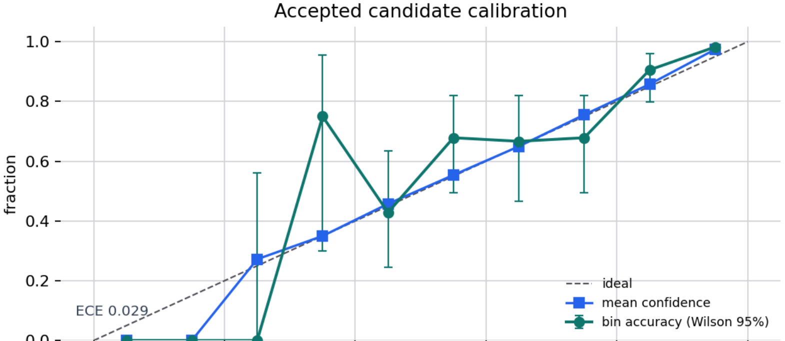
Diagnostic Analysis I

The selected candidate macro F1 is `89.4%`, with a held-out accuracy interval of `86.4% to 91.8%`. Probability diagnostics report expected calibration error `2.9%` and `11` high-confidence error(s). The top non-diagonal confusion pair is `4 -> 9 (4)`, and the minimum selected-candidate accuracy across deterministic robustness transforms is `80.0%`. These values are descriptive diagnostics for the validated local run, not external benchmark claims.

Diagnostic Analysis II

Classification And Error Structure

The class-level and error-pattern diagnostics are intentionally visual-first: @fig:ml_calibration_reliability checks confidence calibration, @fig:ml_classification_metrics_heatmap separates precision, recall, and F1 by digit, and @fig:ml_confusion_pairs ranks the non-diagonal confusions that most shape the selected-candidate error profile. The accompanying tables preserve the same values for audit and downstream comparison.



Candidate Ledger I

Table 15: Candidate lifecycle ledger from `output/data/ml_candidate_ledger.json`.

Candidate	Model	Status	Test accuracy	Parameters
baseline	nearest-centroid	baseline	82.6%	7840
softmax linear	softmax regression	rejected	88.2%	7850
mlp relu 32	MLP	rejected	88.6%	25450
tiny patch attention	tiny patch-attention	rejected	30.4%	5994
mlp tanh 64	MLP	accepted	89.4%	50890
mlp relu 64 deferred	MLP	deferred	N/A	0

Candidate Ledger II

The candidate-selection audit separates the objective ranking from descriptive diagnostics. It records the configured metric, Wilson interval, probability quality, parameter count, and deterministic tie-break context for each evaluated candidate. The diagnostic-boundary table states what each generated surface supports and what it does not support.

Table 16: Candidate-selection audit from `output/data/ml_candidate_selection_audit.json`; the objective metric ranks candidates, while probability diagnostics and parameter counts audit the chosen tie-break context.

Rank	Candidate	Status	Accuracy	Wilson 95%	Brier	NLL	Parameters
1	mlp tanh 64	accepted	89.4%	86.4% to 91.8%	0.161	0.361	50890
2	mlp relu 32	rejected	88.6%	85.5% to 91.1%	0.164	0.380	25450
3	softmax linear	rejected	88.2%	85.1% to 90.7%	0.173	0.390	7850
4	tiny patch attention	rejected	30.4%	26.5% to 34.6%	0.873	2.179	5994

Candidate Ledger III

Table 17: Diagnostic claim-boundary table from `output/data/ml_diagnostic_boundary.json`.

Surface	Source	Method	Supports	Does not support
objective selection	task results	rank evaluated candidates by configured held-out metric and deterministic tie…	accepted-candidate selection within this fixed local task	full MNIST state-of-the-art, external benchmark leadership, or universal mode…
descriptive diagnostics	class diagnostics	per-class metrics, calibration, probability quality, and paired comparison	local error analysis and uncertainty description	population-level certification or deployment readiness
robustness smoke test	robustness report	deterministic no-retrain transforms applied to the fixed test split	small perturbation smoke-test evidence	adversarial robustness or distribution-shift robustness
artifact integrity	integrity attestation	local SHA-256 checks over declared inputs and generated artifacts	local artifact integrity evidence for the run	external signing, production SLSA compliance, or runtime intrusion detection
review governance	review decisions	deferred generated review decisions with human review required	readiness for human review	machine self-approval or publication acceptance

Readiness And Review Artifacts I

The broader AutoResearch run writes the reproducibility, benchmark, review, and manuscript-hydration surfaces summarized below. The schema manifest records 29 schema-versioned governance payload(s), and the local research-object manifest records 81 observed artifact record(s) with checksums and approval state `false`. The phase ledger records 3 settlement pass(es), while the figure-quality report covers 25 registered figure(s) with validity `false`.

Readiness And Review Artifacts II

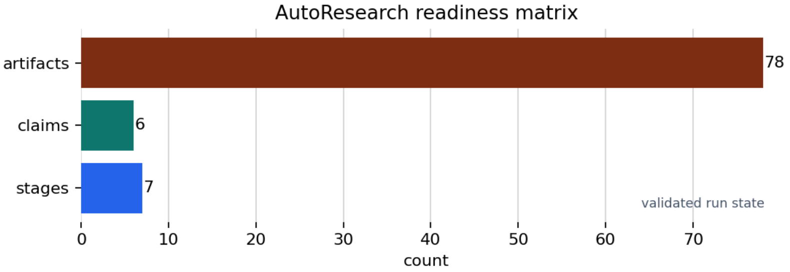


Figure 20: Validated AutoResearch run with 7 stages, 6 supported claims, and 78 required artifacts from output/data/autoresearch_loop.json; the count summarizes readiness artifacts, not human approval. Generation method: Horizontal count summary from final loop metrics. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Readiness And Review Artifacts III

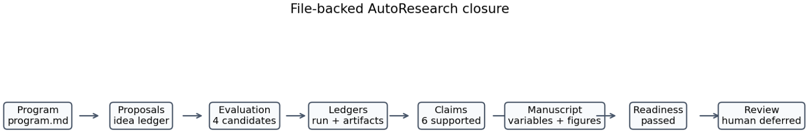


Figure 21: File-backed AutoResearch closure from program through review, with 6 supported claims and readiness passed; review remains a deferred human decision and the provenance path remains inspectable. Generation method: File-backed process-flow diagram from final loop state. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Readiness And Review Artifacts IV

Table 18: Generated artifact manifest from `output/reports/artifact_manifest.json`.

Artifact	Role	Bytes
output/data/autoresearch_claims.json	Loop artifact	1766
output/data/autoresearch_integrity_attestation.json	Security evidence	21905
output/data/autoresearch_inventory_export.json	Security evidence	19464
output/data/autoresearch_loop.json	Loop artifact	15901
output/data/autoresearch_phase_ledger.json	Run or candidate ledger	3779
output/data/autoresearch_plan.json	Loop artifact	17222
output/data/autoresearch_review_packet.json	Review packet	16020
output/data/autoresearch_schema_manifest.json	Loop artifact	6959
output/data/autoresearch_security_profile.json	Security evidence	1537
output/data/autoresearch_stage_matrix.csv	Loop artifact	749
output/data/autoresearch_supply_chain_inventory.json	Security evidence	20683
output/data/autoresearch_threat_model.json	Security evidence	6370
output/data/benchmark_scores.json	Benchmark grading	621
output/data/figure_quality_report.json	Loop artifact	16040
output/data/figure_style.json	Loop artifact	1117
output/data/idea_ledger.json	Run or candidate ledger	5113
output/data/manuscript_figure_blocks.json	Manuscript hydration	13062
output/data/manuscript_variable_provenance.json	Manuscript hydration	30625
output/data/manuscript_variables.json	Manuscript hydration	54579
output/data/ml_bootstrap_intervals.json	Loop artifact	615
output/data/ml_calibration_bin_intervals.json	Loop artifact	2879
output/data/ml_calibration_report.json	Loop artifact	2201
output/data/ml_candidate_intervals.json	Loop artifact	1577
output/data/ml_candidate_ledger.json	Run or candidate ledger	570872
output/data/ml_candidate_rank_stability.json	Loop artifact	3135
output/data/ml_candidate_selection_audit.json	Loop artifact	2145
output/data/ml_class_balance.json	Loop artifact	2393
output/data/ml_classification_diagnostics.json	Loop artifact	4246
output/data/ml_confusion_matrix.csv	Loop artifact	271
output/data/ml_diagnostic_boundary.json	Loop artifact	2064

Readiness And Review Artifacts V

output/data/ml_error_examples.json	Loop artifact	989
output/data/ml_paired_comparison.json	Loop artifact	470
output/data/ml_prediction_records.json	Loop artifact	989913
output/data/ml_probability_diagnostics.json	Loop artifact	3045
output/data/ml_robustness_report.json	Loop artifact	3758
output/data/ml_statistical_summary.json	Loop artifact	3032
output/data/ml_task_results.json	Loop artifact	703566
output/data/ml_training_diagnostics.json	Loop artifact	2968
output/data/ml_training_history.csv	Loop artifact	6775
output/data/mnist_task_config.json	Loop artifact	3926
output/data/research_object_manifest.json	Loop artifact	20107
output/data/research_program.json	Loop artifact	935
output/data/review_decisions.json	Review packet	669
output/data/run_ledger.json	Run or candidate ledger	328
output/figures/autoresearch_candidate_lifecycle.png	Generated figure	28053
output/figures/autoresearch_closure_flow.png	Generated figure	40410
output/figures/autoresearch_integrity_chain.png	Generated figure	46980
output/figures/autoresearch_security_control_matrix.png	Generated figure	86574
output/figures/ml_bootstrap_intervals.png	Generated figure	21834
output/figures/ml_calibration_reliability.png	Generated figure	73889
output/figures/ml_candidate_rank_stability.png	Generated figure	43692
output/figures/ml_candidate_scores.png	Generated figure	59953
output/figures/ml_classification_metrics_heatmap.png	Generated figure	55092
output/figures/ml_complexity_accuracy.png	Generated figure	35135
output/figures/ml_confusion_matrix.png	Generated figure	65820
output/figures/ml_confusion_pairs.png	Generated figure	33982
output/figures/ml_generalization_gap.png	Generated figure	48512
output/figures/ml_learning_curves.png	Generated figure	59258
output/figures/ml_paired_correctness.png	Generated figure	43309
output/figures/ml_per_class_accuracy.png	Generated figure	34800
output/figures/ml_probability_margin_distribution.png	Generated figure	41754
output/figures/ml_probability_quality.png	Generated figure	38071
output/figures/ml_robustness_matrix.png	Generated figure	51332
output/figures/ml_selective_accuracy.png	Generated figure	48274
output/figures/ml_training_dynamics.png	Generated figure	51004
output/figures/mnist_class_balance.png	Generated figure	27059

Readiness And Review Artifacts VI

output/figures/mnist_error_examples.png	Generated figure	28431
output/figures/mnist_subset_contact_sheet.png	Generated figure	23617
output/reports/autoresearch_loop.json	Loop artifact	15901
output/reports/autoresearch_review_packet.md	Review packet	912
output/reports/autoresearch_security_review.md	Review packet	1104
output/reports/autoresearch_summary.md	Loop artifact	255
output/reports/benchmark_readiness_smoke.json	Benchmark grading	778
output/reports/ml_benchmark_score.json	Benchmark grading	382
output/reports/ml_experiment_report.md	Loop artifact	1687

Readiness And Review Artifacts VII

Table 19: Review-gate decisions from `output/data/review_decisions.json`.

Gate	Required	Decision	Rationale
proposal_review	True	deferred	Decision is read from human_review.yaml when present; generated readiness is not approval.
evidence_review	True	deferred	Decision is read from human_review.yaml when present; generated readiness is not approval.

Readiness And Review Artifacts VIII

Table 20: Benchmark grading table from `output/data/benchmark_scores.json`.

Benchmark task	Status	Score	Grading output
readiness-smoke	graded	1	output/reports/benchmark_readiness_
ml-loop-score	graded	1	output/reports/ml_benchmark_score.js

Readiness And Review Artifacts IX

Table 21: Deterministic phase ledger from `output/data/autoresearch_phase_ledger.json`; settlement order is not an autonomy claim.

Phase	Order	Group	Observed artifacts	Description
intrinsic readiness	1	readiness	3	validate configured project-intrinsic contracts
core artifacts	2	loop	0	write plan, stage matrix, and provisional loop outputs
evidence registry	3	evidence	2	write local evidence registry
ml task	4	ml	54	run fixed-seed bounded candidate evaluation
method contract	5	governance	3	write program, idea, run, review, and benchmark ledgers
provisional payloads	6	settlement	12	refresh loop payloads before extrinsic validation
security artifacts	7	security	10	write local security and integrity evidence
final visuals	8	figures	54	write final registry-backed figures
manuscript hydration	9	manuscript	9	write variables, provenance, and figure blocks
readiness manifest	10	settlement	12	refresh checksum manifest before extrinsic validation
schema manifest	11	schema	2	write generated JSON schema-version manifest

Readiness And Review Artifacts X

research object manifest	12	packaging	2	write local research-object manifest
extrinsic readiness	13	readiness	3	validate generated artifacts and extrinsic contracts
final schema manifest	14	schema	2	refresh schema manifest after final payload updates
final research object manifest	15	packaging	2	refresh local research-object manifest
artifact manifest	16	settlement	12	write final artifact checksum manifest

Readiness And Review Artifacts XI

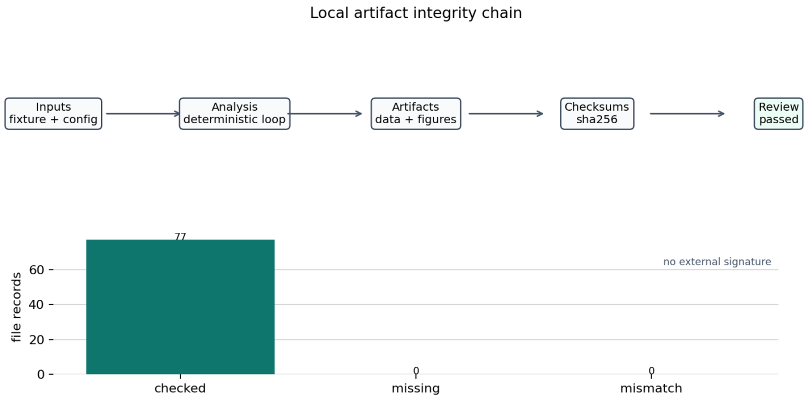
Table 22: Figure-quality checks from `output/data/figure_quality_report.json`; 25 registered figure(s) were checked.

Figure	Pixels	Variance	Source exists	Nonblank
fig:autoresearch_candidate_lifecycle	1184x480	0.083	True	True
fig:autoresearch_closure_flow	1664x448	0.015	True	True
fig:autoresearch_integrity_chain	1440x734	0.040	True	True
fig:autoresearch_security_control_matrix	1470x734	0.014	True	True
fig:autoresearch_stage_matrix	1120x416	0.080	True	True
fig:ml_bootstrap_intervals	1152x448	0.009	True	True
fig:ml_calibration_reliability	1152x832	0.015	True	True
fig:ml_candidate_rank_stability	1408x608	0.053	True	True
fig:ml_candidate_scores	1376x688	0.013	True	True
fig:ml_classification_metrics_heatmap	928x832	0.092	True	True
fig:ml_complexity_accuracy	1120x608	0.010	True	True
fig:ml_confusion_matrix	896x768	0.046	True	True
fig:ml_confusion_pairs	1152x576	0.129	True	True
fig:ml_generalization_gap	1184x864	0.059	True	True
fig:ml_learning_curves	1216x608	0.018	True	True
fig:ml_paired_correctness	768x672	0.075	True	True
fig:ml_per_class_accuracy	1152x512	0.098	True	True
fig:ml_probability_margin_distribution	1184x864	0.024	True	True
fig:ml_probability_quality	1344x608	0.046	True	True
fig:ml_robustness_matrix	1280x608	0.073	True	True
fig:ml_selective_accuracy	1088x608	0.016	True	True
fig:ml_training_dynamics	1408x608	0.072	True	True
fig:mnist_class_balance	1216x544	0.071	True	True
fig:mnist_error_examples	1280x734	0.234	True	True
fig:mnist_subset_contact_sheet	1216x544	0.228	True	True

Security Readiness And Integrity Evidence I

The local security profile reports attestation status `passed` after checking 77 file record(s), with 0 missing record(s) and 0 checksum mismatch(es). The inventory contains 14 input record(s) and 66 generated-artifact record(s). The integrity-chain figure is @fig:autoresearch_integrity_chain. These values support local artifact-integrity claims only; they do not claim external signing, production SLSA compliance, or runtime security monitoring.

Security Readiness And Integrity Evidence II



Security Readiness And Integrity Evidence III

Figure 22: Local integrity chain from output/data/autoresearch_integrity_attestation.json; checksums summarize the observed run artifacts and remain unsigned local evidence. Generation method: Local checksum attestation chain with checked, missing, and mismatch counts. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Table 23: Integrity-attestation summary from `output/data/autoresearch_integrity_attestation.json`.

Integrity field	Value
status	passed
algorithm	sha256
checked files	77
missing files	0
checksum mismatches	0
external signature	False

Manuscript Hydration Provenance I

The final run supports 6 manuscript-facing claim(s) and checks 78 required artifact(s). The rendered manuscript uses injected variables from generated data payloads, so the abstract and results track the latest analysis run rather than hard-coded counts. The final readiness status is `passed`; generated review decisions are recorded as deferred for human review rather than as self-approval.

Table 24: Variable provenance summary from `output/data/manuscript_variable_provenance.json`.

Source artifact	Injected variables or fragments
output/data/autoresearch_integrity_attestation.json	5
output/data/autoresearch_loop.json	55
output/data/autoresearch_phase_ledger.json	2
output/data/autoresearch_schema_manifest.json	1
output/data/autoresearch_security_profile.json	6
output/data/autoresearch_supply_chain_inventory.json	3
output/data/autoresearch_threat_model.json	4
output/data/benchmark_scores.json	1
output/data/figure_quality_report.json	3
output/data/manuscript_variable_provenance.json	1
output/data/ml_bootstrap_intervals.json	3
output/data/ml_calibration_bin_intervals.json	1
output/data/ml_calibration_report.json	3
output/data/ml_candidate_intervals.json	1

Manuscript Hydration Provenance II

output/data/ml_candidate_ledger.json	1
output/data/ml_candidate_rank_stability.json	3
output/data/ml_candidate_selection_audit.json	1
output/data/ml_class_balance.json	3
output/data/ml_classification_diagnostics.json	5
output/data/ml_diagnostic_boundary.json	1
output/data/ml_paired_comparison.json	3
output/data/ml_probability_diagnostics.json	4
output/data/ml_robustness_report.json	2
output/data/ml_statistical_summary.json	9
output/data/ml_task_results.json	39
output/data/ml_training_diagnostics.json	7
output/data/research_object_manifest.json	2
output/data/review_decisions.json	1
output/figures/figure_registry.json	51
output/reports/artifact_manifest.json	1